What can public fatigue data actually buy?

A controlled training-expansion and few-test stress-selection study

Working addendum to the phase 3.1 paper. Human authorship and affiliations remain to be supplied. The human separately requested author data; this local workflow sent no correspondence or submission.

Abstract

We expanded a source-reviewed LPBF316L fatigue training cohort from 309 to 441 observations, while retaining publication reuse blocks and right censoring. A new 2025 numerical table supplied 69 separate experimental outcomes. On its 25 acquisition-eligible observations, additional training data reduced zero-shot hierarchical lower-life violations from 57.5% to 46.7%; after three random local tests both training regimes gave 20.0%. A separate 115-observation, two-family holdout replay showed that selecting high stresses cut observed support cycles by 70.6%, but lowered predicted q10 life relative to random selection to 0.872 × for hierarchical Basquin and 0.688 × for a random-fatigue-limit model. These results support a reproducible open-data benchmark and a cost–utility counterexample; they do not establish a universally better acquisition policy or certified design reliability.

Collected evidence, with non-additive counts

Asset	Evidence and use
441 training observations	48 conditions; 14 DOIs; 13 source-family blocks; 380 failures+61 censors. 132 newly source-qualified coordinates were already in the public database.
69 new numerical outcomes	64 failures+5 runouts; 14 conditions; one 2025 author program, held out. Publication test numbers are not laboratory/build IDs.
20 author-ID LCF/SWT lives	Separate multiaxial/strain-controlled subset plus 200 defect measurements; not prescribed-stress HCF data.
7 interrupted specimens	217,368 force/strain records in peak-valley and selected-loop views of the same 7 specimens; zero Nf rows manufactured.
3 additional cyclic histories	14,499 usable cyclic-response rows; 115 orphan rows retained. Same Jambor family conservatively.
2 selected AM-stainless traces	1,087,236 time samples; grade/temperature unresolved. No fabricated 316 L label.

Counts across related repositories and different measurement types must not be summed as independent fatigue specimens. Four public raw packets were acquired fully or selectively; one large archive was inventoried by public byte ranges. A separate 2026 rotating-bending table adds 67 reported points, including 25 reused points and 42 from the other three conditions; it is not used in these frozen axial fits.

Controlled training expansion

The comparison uses identical test queries and revealed supports. Four predeclared populations are 309, 441, 347(309+Elanges 38) and 403(309+Zhang 94). The same 309-trained planner chooses supports for every predictor and population. Six unchanged censored statistical/physical baselines and four fixed acquisition policies are evaluated at ten hash seeds. Source-balanced likelihood weights change when families are added; this estimates a controlled evidence expansion, not an isolated causal effect of row count or diversity.

The new Yankin 2025 table reports stress amplitude and individual failure lives directly. Five nominal 1e6 censor stops are derived from plots and the stated life range. Test temperature, explicit feedback mode, Kt, laboratory specimen IDs and build/powder identities are not supplied. Therefore this is a temperature-unstated source panel, separate from the identity-qualified room-temperature benchmark.

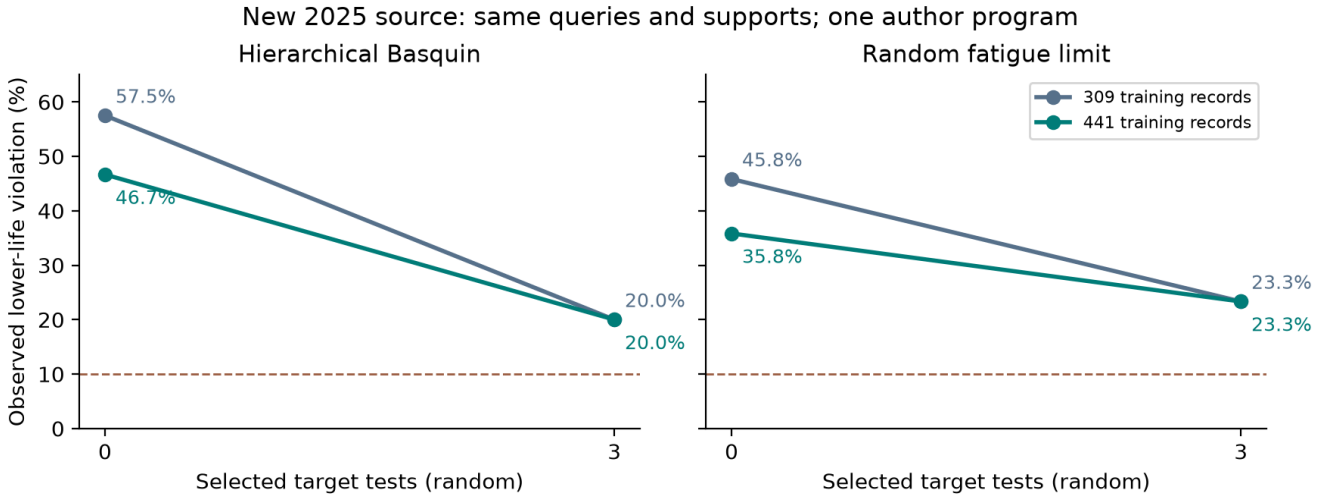


Figure 1. Source/condition-balanced violation fractions, averaged over shared queries at ten repeated splits. One author program; no population confidence intervals. Dashed line: nominal 10% risk target, not a guarantee.

Interpretation

At k0 , more verified training evidence makes predictions more conservative and reduces observed violations, but risk remains much higher than 10%. At k3 the hierarchical B/A life ratio is 0.998 and violations are unchanged at 20.0%; RFL life ratio is 0.949 with violations unchanged at 23.3%. Data quantity alone does not fix the new-source error.

Only four conditions/25 observations allow the inherited query/pool rule and k3 . Three pools contain exactly three candidates, so k3 exhausts them and policy choice has no set-level effect. The remaining 44 observations are retained, not discarded as invalid. No k5 / k10 result is invented.

Where should the few tests be placed?

The separate source-family extension uses 115 eligible observations/11 conditions from the 132 source-qualified coordinates. When Zhang is queried, training B is 347; when Elanges is queried, B is 403. The entire target family is absent from both training regimes. This extension was frozen after the main fits, before their numerical summaries were inspected, in response to the pool-size limitation; it is explicitly exploratory.

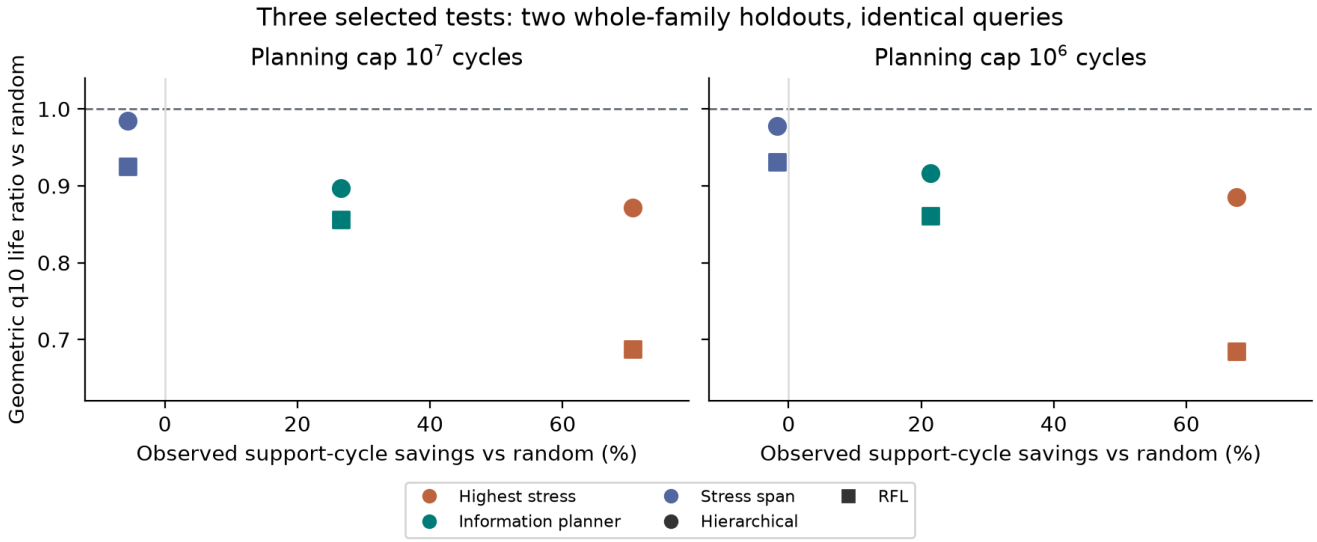


Figure 2. Same-query k3 comparisons versus random, source-family-balanced. Circle: hierarchical; square:RFL. Costs are observed replay cycles, not wall-clock or new experimental costs. Negative savings mean more cycles.

Policy at C 1e7 , k3	Cycle savings	Life: hier / RFL
High stress	70.6%	0.872 × /0.688 ×
Information planner	26.6%	0.897 × /0.857 ×
Stress span	−5.6%	0.985 × /0.925 ×

At C 1e6 , high-stress savings are 67.5%, with life ratios 0.885/0.685. The active planner is not uniformly superior. All six baselines, k0 /3/5 where feasible, risk differences, availability and two-family descriptive intervals are in the CSV supplement; displayed models are representative physical/statistical controls, not winners selected for superiority.

Random k3 adaptation itself has model-dependent value: hierarchical life improves 1.89 × over its matched k0 while observed violations fall from 12.5% to 3.6%; RFL life improves 1.33 × but violations rise from 1.25% to 2.56%. Larger lower-life predictions alone are not evidence of a better decision.

Reproducibility and contribution boundary

Provenance

Three admitted publications supply the 132-coordinate increment, blocked into two families. A whole 11-point reused horizontal curve is excluded; 14 censor coordinates are replaced by stated 1e6 / 2e6 stops, preserving original values. Unverified database 25°C/Kt 1 values are removed from certified fields. No fitted curves, tensile means or caption anchors become extra specimens. Independent source, schema and arithmetic reviews pass.

Validation

The four experiments produce 164,160 prediction instances and 5,200 selected-support instances, not that many independent tests. Independent reviews reconstruct summaries, right-censor arithmetic, family exclusions, queries and support costs. All 16 unique population fits converge without fallback or selected bound hits. A fresh isolated replay reproduces 84 scientific CSVs byte for byte and the fitted parameters exactly. Nineteen synthetic model/design tests pass. Across 1,156 fixed-parameter RFL checks,96/192-node q10 differences are at most 4.78e-10 log 10 cycles, without numerical exceptions or infinity-classification changes.

What is defensible

The contribution is a curated open proxy dataset and a controlled benchmark showing when extra source evidence or cheap high-stress tests help, fail or trade life utility for cost. It is not a new named acquisition algorithm, exact Centola replication or a demonstrated improvement over the original BNN. The 441 cohort has DOI overlap 8/17 with Centola; neither point identity nor original risk volume/SWT/Owen targets is established.

Remaining scientific work

The Mérot physical-lineage panel remains valid and separate. A second fully mapped independent room-temperature author campaign is still missing. More source-qualified data can continue to be collected and tested while that lineage question remains open. Stronger policy evidence needs multiple independent programs with candidate pools larger than the chosen test budget, complete endpoint/method fields, and successful paired safety-and-utility tests. Two-family bootstrap intervals and repeated seeds do not supply that evidence.

Primary sources and local evidence

Centola et al.; original 518 rows on request — https://doi.org/10.1016/j.engfailanal.2024.109081

FatigueData-AM 2022; licensed source coordinates — https://doi.org/10.6084/m9.figshare.22337629

Zhang et al.; 2017 original experiments — https://doi.org/10.1016/j.msea.2017.07.071

Zhang et al.; 2019 HCF/ratcheting experiments — https://doi.org/10.1016/j.ijfatigue.2018.12.016

Elangeswaran et al.; original CA post-treatment study — https://doi.org/10.1016/j.ijfatigue.2019.01.013

Yankin,Perveen,Talamona; Table 3 — https://doi.org/10.1038/s41598-025-89003-2

Bartosak multiaxial LCF/SWT raw data — https://zenodo.org/records/12515773

Jambor interrupted raw histories — https://zenodo.org/records/14763342

Jambor cyclic-softening raw data — https://zenodo.org/records/17985589

AM/wrought-stainless raw traces — https://zenodo.org/records/17086687

Separate 2026 rotating-bending table; reuse linked — https://doi.org/10.1016/j.ijfatigue.2025.109365